Aarush Chaudhary

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Education

SVKM's Narsee Monjee Institute of Management Studies — Hyderabad, India
B. Tech, Computer Science & Engineering (Data Science) — *2024 – 2028 (Expected)*

- **Coursework**: Data Structures & Algorithms, Distributed Systems, Object-Oriented Programming, DBMS, Machine Learning, Artificial Intelligence, Operations Research

Experience

Technical Lead — Hyderabad, India
CodeIT Coding Club — *July 2025 – Present*

- Collaborated across cross-functional teams to design, architect, and deploy full-stack digital workflows, scaling event logistics and student engagement.
- Fostered a culture of software engineering excellence by mentoring peers in data structures, algorithms, and complex system design.

Founder & Chairperson — Hyderabad, India
IEEE Student Branch, SVKM's NMIMS — *December 2025 – Present*

- Established the university's first IEEE Student Branch, driving community building and onboarding 15 founding members to accelerate technical research.
- Structured the administrative and technical framework to support large-scale collaborative engineering projects and AI research initiatives.

Student Network Team Member — Hyderabad, India
IEEE Hyderabad Section — *February 2026 – Present*

- Collaborated within the regional section network to coordinate section-level technical initiatives, driving student engagement and fostering a culture of engineering excellence.

Projects

NMPralekh: Distributed MIS Dashboard | *React, Django, PostgreSQL, Redis, Celery, Podman*

- Architected a highly scalable, distributed Management Information System deployed across 9 university campuses to process massive datasets.
- Engineered the backend using Django and PostgreSQL, utilizing Redis caching and pgBouncer to achieve sub-50ms API response times under heavy load.
- Implemented concurrency and asynchronous task execution using Celery and Gunicorn, containerizing the complex system with Podman for production reliability.

NMIMS Campus Assistant (AI-Integrated Software) | *Python, Flask, AWS, FAISS*

- Designed and implemented an AI-integrated federated RAG system with isolated vector stores to parse and query complex, large-scale institutional datasets.
- Built a hybrid query router (Regex + LLM) to classify user intent and optimize data retrieval performance across multiple domain indexes.
- Engineered a live-reload data pipeline using AWS Textract and S3 to asynchronously process documents and maintain the system's infrastructure.

BioShield IoT Security Network | *Python (FastAPI), Kotlin, React, Redis*

- Developed a high-performance, end-to-end IoT security infrastructure utilizing a Python backend, Redis caching tier, and React web interface.
- Implemented multi-threaded biometric data processing workflows and custom cryptographic hashing for real-time, highly reliable authentication.
- Designed a robust security layer featuring a dynamic backend key vault and granular Role-Based Access Controls (RBAC) for production use.

Afterlife Protocol | *Next.js, Python, Algorand, AWS (RDS, S3, EC2)*

- Designed a complex, decentralized storage and inheritance system utilizing Algorand Smart Contracts for a trustless multi-sig execution flow.
- Built a high-performance Python Oracle to monitor network transactions and synchronize distributed application states with a PostgreSQL database.

Achievements

- **Winners** – AI4Health Project Expo at SVKM's NMIMS, STME, Hyderabad
- **Runner Up** – Nexathon 24-Hour Hackathon on Algorand Blockchain in Open Innovation at Symbiosis Institute of Technology
- **IoT Track Winners** – Innovathon 24-Hour Hackathon at SVKM's NMIMS, STME, Indore

Publications

Campus Assistant: A Scalable Conversational AI using Domain-Specific Federated RAG

- **Published**: 2026 IEEE International Conference on Emerging Computing and Intelligent Technologies
- Co-authored research proposing a Federated RAG architecture to solve "context collision" in large-scale data environments.
- Conducted rigorous systems data analysis, demonstrating a 23% increase in response accuracy and an 11% reduction in hallucinations compared to baselines.

Technical Skills

Languages: Python, C/C++, JavaScript, SQL, R, HTML/CSS
Data, AI & Core CS: Data Structures & Algorithms, Distributed Systems, Artificial Intelligence, Machine Learning, Computer Vision, RAG architectures
Cloud, Tools & Frameworks: Git, Linux, AWS (S3, RDS, EC2, Bedrock), Docker, Podman, Redis, Node.js, Next.js, Django, FastAPI